Advanced Mathematical Techniques Assignments
Dr. Osborne 2026
G. Arfken and H. Weber, *Mathematical Methods for Physicists*, Elsevier Press, 6th. Ed, 2005.
* means possible error in book

Assn	Topic(s)	Assignment	Start	Due
	First Day Sheet due 22 August		2/2	2/6
1	Functional Representations Power Series	5.1.1; 5.2.7, 10; 5.6.14, 17 Worksheet	2/9	2/20
Q1	Sequences & Series Convergence Taylor Series Integration Techniques	Quiz 1 on 19 February		
2	Expansions Approximation Techniques Infinite Products	5.7.4, 7, 15, 19(a); 5.8.1; 5.11.5, 6 Worksheet	2/10	2/25
3	The Gamma Function	5.9.13; 8.1.7, 8 Worksheet	2/19	2/27
Q2	Infinite Products Gamma Function Properties	Quiz 2 on 2 March		
4	The Beta Function Integral Representations Special Integrals Integrals including Logarithms	5.9.12 (just exact, not series); 8.4.6, 10 Worksheet	2/24	3/6
5	More Integrals including logarithms Beta Function take 2	Worksheet	2/26	3/11
6	Beta Function take 3 Regularization of Singularities	Worksheet	3/3	3/13
Q3	Beta Function Representations Regularization of Integrals	Quiz 3 on 12 March		
7	More Regularization The Riemann Zeta Function	5.9.8, 9; 8.2.14 Worksheet	3/5	3/19
8	Asymptotic Expansions Reduction to Quadrature First and Second Integrals; Periods	5.10.4 (hint: $u = t^2$), 8*; 8.4.14, 15; 9.2.5,14 Worksheet	3/12	3/27
Q4	Regularization Zeta Function Asymptotic Expansions	Quiz 4 on 24 March		
9	Mathematical Models Probability Distributions	8.1.23, 24; 8.4.17 Worksheet	3/19	4/15
Q5	Probability Distributions Multinomial Distribution	Quiz 5 on 14 April		
10	Thermodynamics	Worksheet	4/9	4/24

11	The Boltzmann Factor Atomic Speed Distributions Planck & Stefan-Boltzmann Law Statistical Mechanics	5.6.19* (geometric!); 8.1.5; 8.2.16 Worksheet	4/14	4/29
Q6	Boltzmann Factor Maxwell-Boltzmann Speed Dist Equipartition of Energy Photons Thermodynamics	Quiz 6 on 28 April		
12	Groups, Rings, and Fields	Worksheet	4/27	5/11
Q7	Groups, Rings, and Fields Basic Linear Algebra Structure	Quiz 7 on 12 May		
13	Linear Algebra	Worksheet	4/30	5/22
	Final Activity on 28 May			